

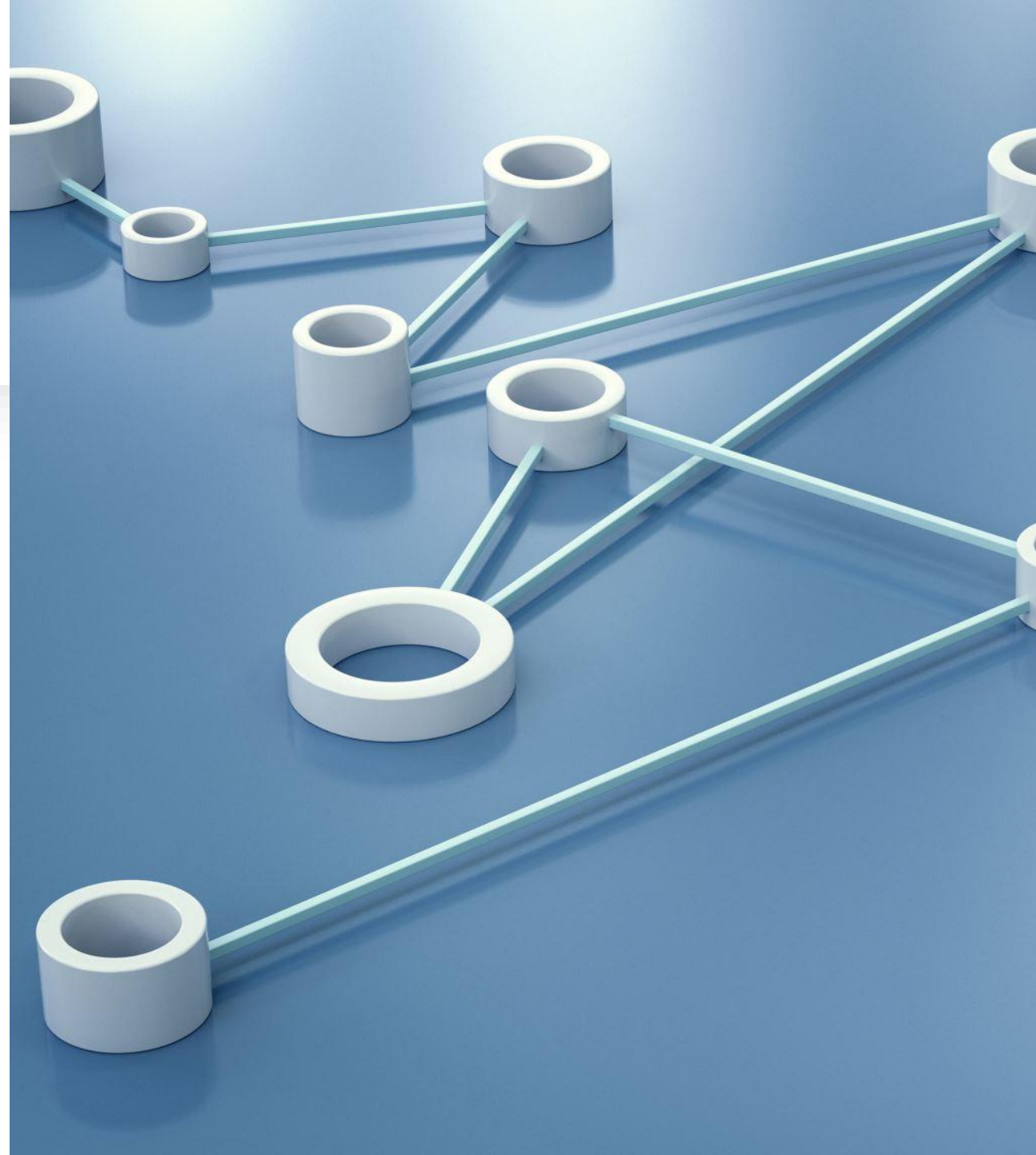
# Networks and Systems Management

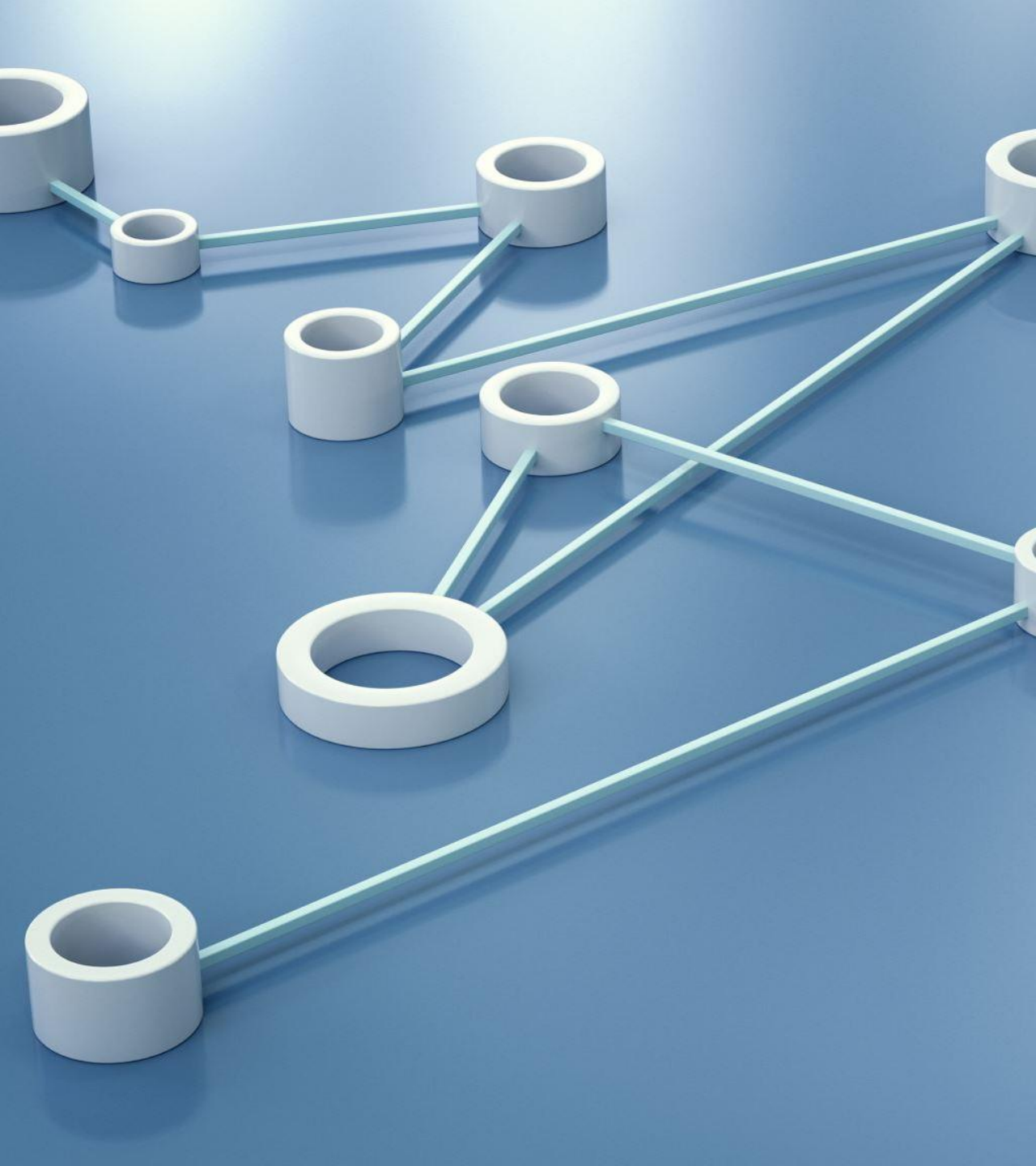


Quality Network  
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# Introduction

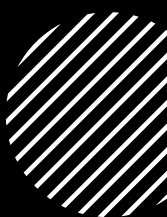
- What are quality networks?
- Importance of network quality in various domains (business, education, healthcare, etc.)
- Overview of key discussion points





# Characteristics of a Quality Network

- High reliability and uptime
- Low latency
- Scalability
- Security and data protection
- Efficient bandwidth utilization



# Importance of Network Quality



Ensuring seamless communication



Supporting high-demand applications (e.g., video conferencing, IoT, cloud computing)



Enhancing user experience



Reducing operational costs



Improving cybersecurity

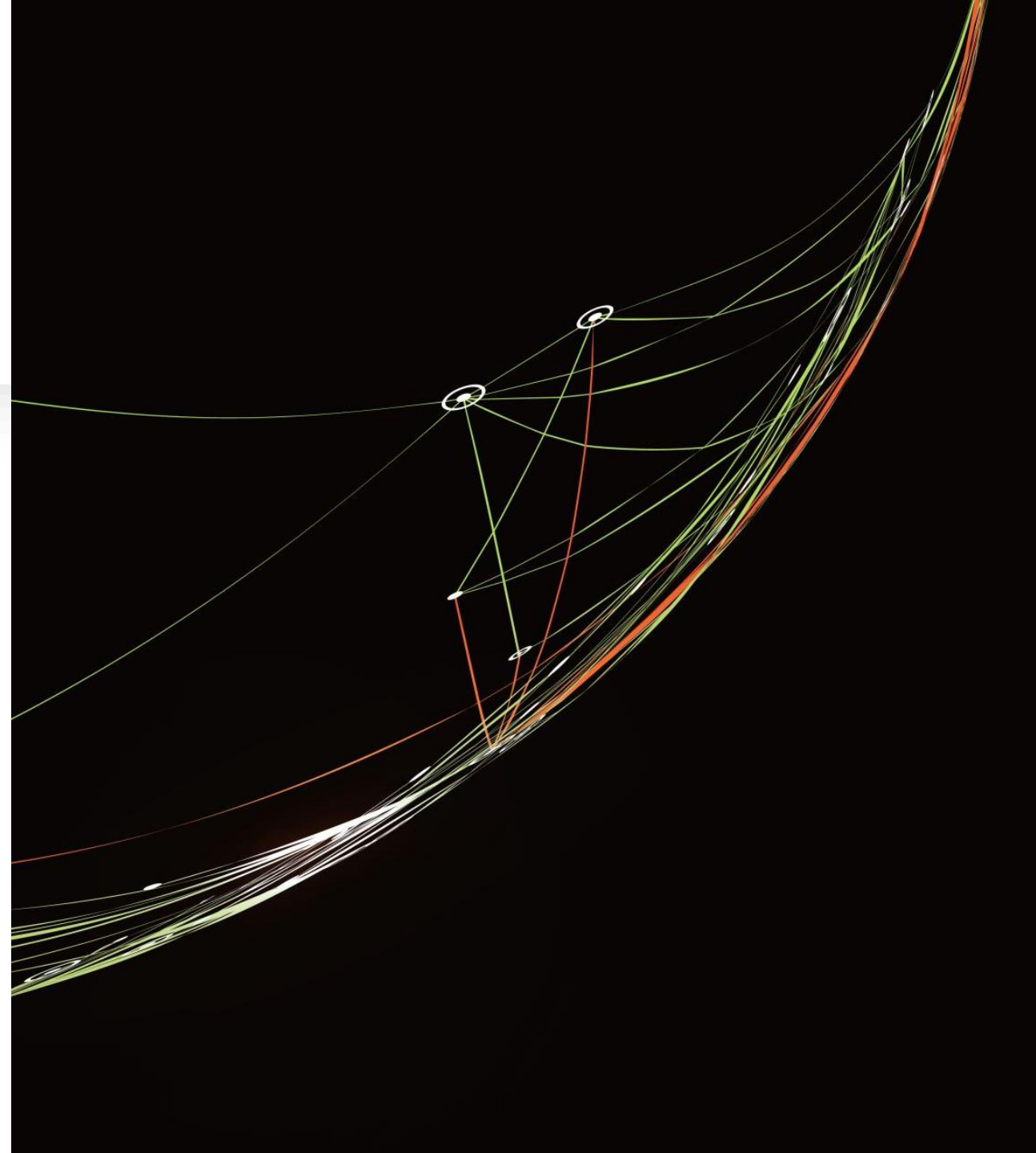
# Challenges

- Large number of devices, and switching limitations
  - Leads to capacity bottlenecks
- Specific requirements for users and applications
  - with service level commitments with users/clients
- Costs (!)
  - both CAPEX and OPEX
- Traffic growth
  - and shrinkage (?)
- Network outages
- Users with different levels of access and accounts
- Attacks



# Network design

- Do you have an idea of how the traffic flows in your network?
  - North-south, east-west, other?
- Different parts of the network may use different technologies and topologies
  - LAN, WAN, MAN ; access, distribution, core
  - STP, fat tree, etc.
  - 1 – 10 – 25 – 40 – 100 Gbps
- Segregation:
  - Workstations, Servers/Datacenter, public-facing (DMZ), admin, etc.
- Technologies:
  - Ethernet, VLAN, EtherChannel, MPLS, IP, OSPF, BGP
- Interconnection with other networks
  - Internet Service Providers (ISP), BGP Peers, other networks of other departments



# The “dev | fence | ops” trap, configuration

- Silos
  - Network Planning and Design
  - Network Deployment
- Devops for networking allow a more iterative process of designing, deploy, and getting feedback to update the network design
- Network function virtualization – helps
- Cost of hardware and hardware compatibility with future network expansions – hinders



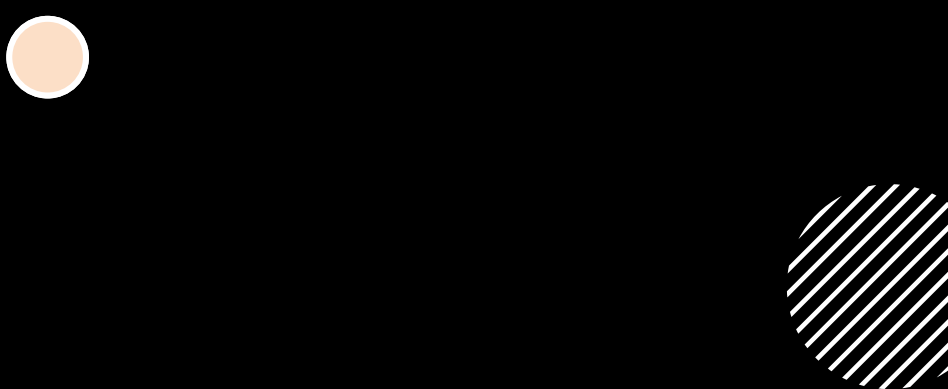
# Application quality requirements

- Capacity, bit/s – bandwidth-intensive applications
  - Bursts – timescale – how long, how many bytes
  - Capacity vs. throughput vs. goodput
- Delay – real-time, interactive applications
  - End-to-end delay (control)
  - Round-trip delay (teleconference)
  - Delay variation/jitter – visualization
- Reliability – mission-critical applications
  - Error rates – bit, packet, etc.
  - Mean time between failures - MTBF
  - Mean time to recover - MTTR
  - $\text{Availability} = \text{MTBF} / (\text{MTBF} + \text{MTTR})$ ,  $\text{Uptime}(\%) = 1 - \text{Availability}$
  - 99.999% uptime (5 nines) ó 5.3 minutes downtime in a year





# QoS and traffic engineering, SLA



## Best effort networks

- Lightly used – quality ok; Heavily used – quality degradation

## Quality of service (QoS)

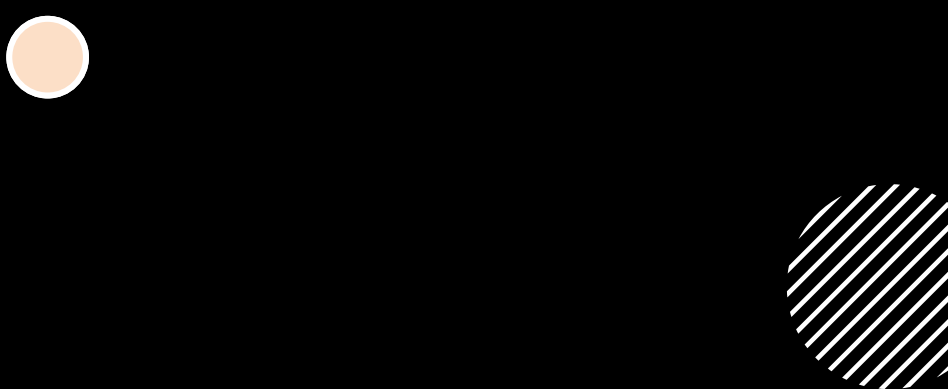
- Queue management – choose packet, different queues
- Round-robin, token bucket, RED, etc. – algorithms

## tc Linux

- <https://www.cyberci8.biz/faq/linux-traffic-shaping-using-tc-to-control-hXp-traffic/>
- Add 200ms delay, add a token bucket
  - `tc qdisc add dev eth0 root netem delay 200ms`
  - `tc qdisc add dev eth0 root tbf rate 1mbit burst 10kb latency 70ms peakrate 2mbit minburst 1540`



# QoS and traffic engineering, SLA



ATM, intserv, diffserv+MPLS



SLA – uptime (%), minimum  
bitrate and delay, etc.



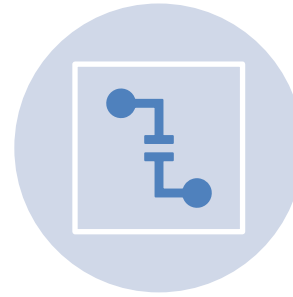
Why is QoS not a  
problem in circuit  
switching networks?

What is  
admission  
control?

# Faults



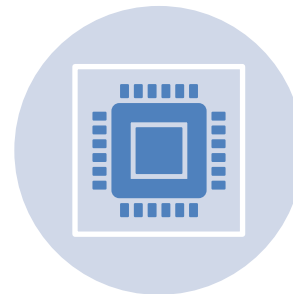
Both hardware and software



Root cause analysis –  
'root cause' detector hard  
in complex networks

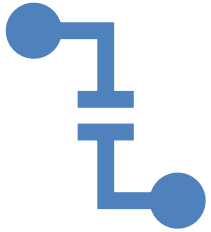


Fault recovery – agile  
reconfiguration



Fault detection – signal  
processing and machine  
learning

# Security



## Enforcing security

Segregation

Access control

Firewall

IDS/IPS

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## Security management

Vulnerability scanning

Intelligence gathering

Incident response

Forensics

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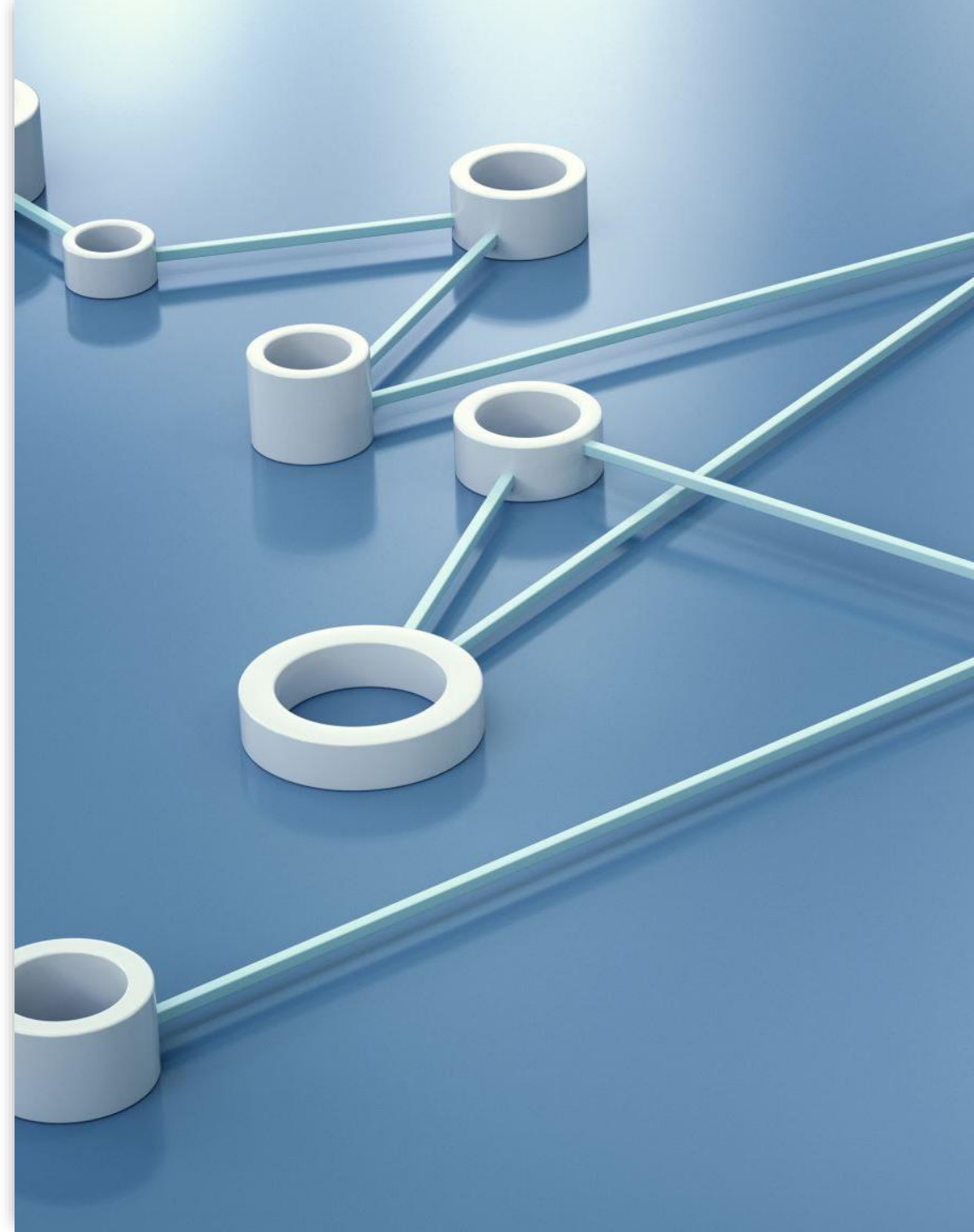
# Emerging Technologies in Quality Networks

- 5G and its impact on network performance
- Software-Defined Networking (SDN)
- Network Function Virtualization (NFV)
- AI and Machine Learning for network optimization
- Edge computing for reducing latency



# Network design (more)

- How do you expose part of your network to the public without isolating that part from the rest of the network?
  - How does a DMZ work?
- What are typical network designs?
  - For a corporate network
    - <https://www.ciscopress.com/articles/article.asp?p=2448489>
  - For a cloud provider
  - For an ISP
    - [https://au.int/sites/default/files/documents/31363-doc-session\\_8-1-isp-network-design.pdf](https://au.int/sites/default/files/documents/31363-doc-session_8-1-isp-network-design.pdf)
- How does the Internet topology look like?
  - What is a PoP?
  - What is an Internet Exchange?
  - What is BGP?



# Key Takeaways

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Quality networks ensure reliability, security, and efficiency

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Investing in infrastructure and monitoring improves performance

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Emerging technologies like 5G and AI are shaping network evolution

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Security and scalability are ongoing challenges requiring proactive measures

# Further Reading & References

- "Computer Networking: A Top-Down Approach" by Kurose & Ross
- IEEE and ACM articles on network quality and performance
- Industry reports on 5G, SDN, and network security
- Websites: Cisco, IEEE Xplore, Gartner reports

